Detecting and Predicting Earthquake Ground-Motion Directionality Differences Between Nearby Sites Using Machine Learning

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UE24CS352A – Machine Learning · Mini-Project | Code (public GitHub repository):
https://github.com/DEEKSHAGURURAJA/earthquake-directionality-ml

1. Problem Statement

During an earthquake the shaking at a site is strongest along one horizontal orientation (the RotD100 orientation). Building codes treat this orientation as uniformly random, but observations show that nearby sites have correlated orientations. Our assigned problem, based on Poulos and Silva-Lopez (Stanford), is to learn a function $\Delta\theta = f(s_i, s_j, E_k)$ that takes the seismological, geological and geographical attributes of two recording sites s_i, s_j and of the earthquake E_k, and predicts the absolute difference $\Delta\theta$ (0–90°) between the orientations of maximum shaking at the two sites. We compare a distance-only benchmark, ridge regression and a neural network.

2. Dataset

We use the NGA-West2 database from PEER: a flatfile with 274 attributes for each of 21,539 records of shallow crustal earthquakes, plus the two-component acceleration time series (.AT2 files). $\Delta\theta$ is not in the flatfile, so it was computed from the waveforms. Following the paper, only earthquakes of magnitude ≥ 5.0 were kept. PEER limits waveform downloads to roughly 200 records per two weeks, so we could download waveforms for 198 records. Stations that recorded the same earthquake were paired (all pairs of different stations); the paper kept pairs ≤ 5 km apart, but only 20 such pairs existed in our data, so we allowed pairs up to 100 km. **Our dataset is therefore far smaller than the paper's (about 5,300 pairs) and our results are not directly comparable with it.**

Stage	Records / examples
NGA-West2 flatfile records	21,539
Magnitude ≥ 5.0	8,494
Records with downloaded waveforms (orientation computed)	198
Site pairs ≤ 100 km (same earthquake)	2,010 pairs from 17 earthquakes
of which pairs ≤ 5 km	20
Train / validation / test (80/10/10)	1,608 / 201 / 201

Each pair has 25 candidate features: inter-site distance; earthquake magnitude, hypocentre depth, fault rupture width and length, dip, rake and mechanism; and for the two sites the mean and absolute difference of epicentral, hypocentral, Joyner-Boore and closest distances, Vs30, PGA, PGV and PGD, plus the pair of NEHRP site classes. Numeric features were standardised (zero mean, unit variance) and categorical ones one-hot encoded.

3. Approach



Figure 1. End-to-end pipeline.

Orientation (RotD100). Both horizontal components of a record are passed through a damped single-degree-of-freedom oscillator (period 1 s, 5% damping); because the oscillator is linear, the response in direction θ is $\cos\theta \cdot u_1 + \sin\theta \cdot u_2$, so the orientation of maximum response is found by scanning 0–179°. Component compass directions are read from the file names (e.g. …180/…270, …NS/…EW) so that all sites share one reference frame. **Target:** $\Delta\theta = |\theta_i - \theta_j|$ (period 180°) folded into [0°, 90°]. **Benchmark:** $\Delta\theta = c_1 + c_2 d$ (distance only). **Ridge regression:** least squares with an L2 penalty α; features chosen by backward ranking (drop in validation R^2 when a feature is removed) then forward selection (stop when validation R^2 falls), alternating with tuning of α on the validation set. **Neural network:** multilayer perceptron (ReLU, Adam) on all features with a grid search over layers {1, 2}, neurons {10, 20} and L2 strength {1, 10}, chosen by validation R^2. Models are compared by R^2 and mean absolute error (MAE) on the held-out test set.

4. Implementation Overview

Python (NumPy, pandas, SciPy, scikit-learn, matplotlib). *waveform.py*: .AT2 reader, oscillator response, RotD100 orientation. *pairs.py*: orientations for all records, site pairing and Δθ. *modeling.py*: split, preprocessing, feature selection, benchmark, ridge and neural network. *run_pipeline.py*: runs everything and saves metrics, figures and trained models; *demo.py*: predicts Δθ for two chosen sites of one earthquake; *make_report.py*: write-up. A fixed random seed (42) makes runs reproducible; the README gives set-up and run instructions.

5. Results

Model	R^2 (train)	R^2 (test)	MAE test (deg)	Settings chosen
Benchmark (distance only)	0.005	0.002	23.0	—
Ridge regression	0.005	0.002	23.0	α = 0.001; 1 feature kept (distance)
Neural network	0.084	0.014	22.9	2 layers × 20 neurons, α = 1

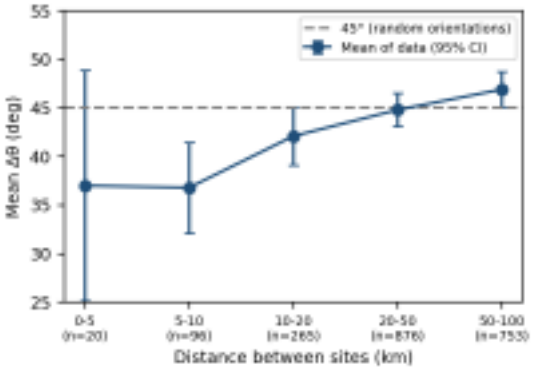


Figure 2. Mean Δθ by distance between sites (final run, 2,010 pairs).

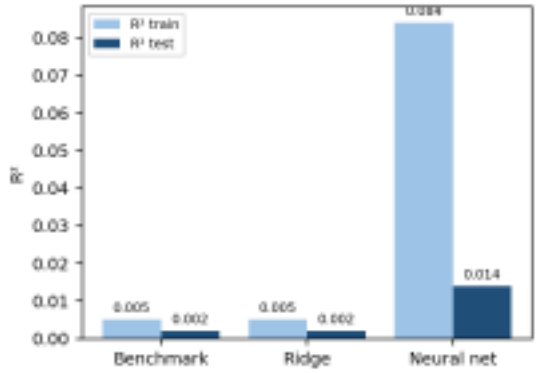


Figure 3. Train vs test R^2 of the three models.

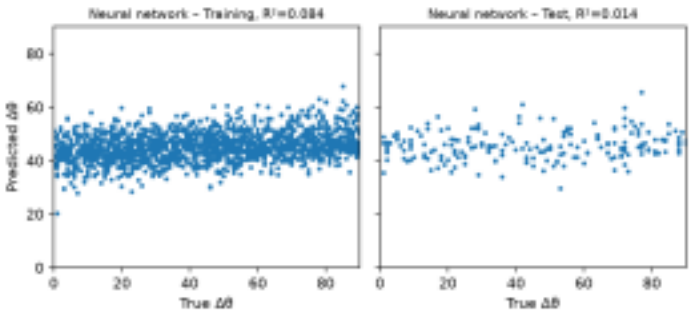


Figure 4. Data vs ridge regression, moving average over distance.

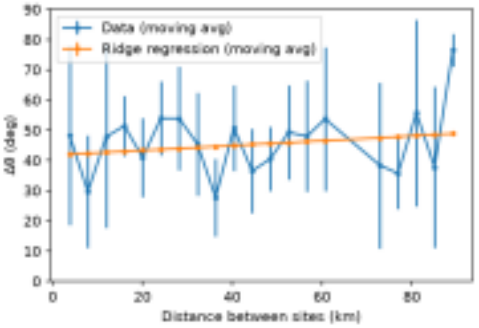


Figure 5. Neural network: true vs predicted Δθ (train, test).

Distance (km)	0–5	5–10	10–20	20–50	50–100
Pairs	20	96	265	876	753
Mean Δθ (deg)	37.0	36.8	42.1	44.8	46.9

Std. dev. (deg)	27.1	23.4	24.8	26.0	26.0

6. Discussion

Trend. Mean $\Delta\theta$ rises with distance (about 37° below 10 km to about 47° at 50–100 km) and approaches the 45° expected for independent random orientations, which supports the paper's finding that orientations are correlated only for nearby sites. **Accuracy.** The test R^2 of all models is close to zero and the MAE (about 23°) is close to 22.5°, the error of always predicting 45°; so the models do not predict individual pairs better than a constant. Ridge selected only the distance and therefore equals the benchmark. The neural network fits the training data better (0.084) than the test data (0.014), a sign of overfitting. **Why the results are weak.** The paper used about 5,300 pairs within 5 km (R^2 0.025, 0.245, 0.354 for the three models); we had 20 such pairs, 17 earthquakes and a 100 km limit where orientations are mostly uncorrelated. Pairs from one earthquake are not independent, so a random split can overstate performance, and with few earthquakes the test scores are unstable.

7. Conclusions and Future Work

We built a complete pipeline, from raw PEER waveforms to RotD100 orientations, site pairs, feature engineering, three regression models and evaluation, and ran it on real data. With the data we could download the models do not beat a constant prediction (test $R^2 \approx 0.00$–0.01, MAE ≈ 23°), but the data show the expected rise of $\Delta\theta$ with distance. With the full NGA-West2 waveform set and the 5 km limit we would expect clearer results; further work includes splitting by earthquake, predicting the mean and variance of $\Delta\theta$, and extending the model to more than two sites.

8. Contributions

Deeksha G and Harshitha B worked together on all parts of the project: data download and processing, code, experiments, write-up and slides.

References. Poulos, A. and Silva-Lopez, R., *Detecting and predicting earthquake ground motion directionality patterns using machine learning tools*, Stanford. Ancheta, T. D. et al. (2014), NGA-West2 database, *Earthquake Spectra* 30(3):989–1005. Boore, D. M. (2010), *Bull. Seismol. Soc. Am.* 100(4):1830–1835. Pedregosa, F. et al. (2011), Scikit-learn, *JMLR* 12:2825–2830.